
References of training material updates #2

NGSOTI Project: 101127921
DIGITAL-ECCC-2022-CYBER-03 D3.2

Team CIRCL/NGSOTI



Co-funded by
the European Union



ECCC 
EUROPEAN CYBERSECURITY
COMPETENCE CENTRE

2026-06-30

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1 References of training material updates #2

1.1 Disclaimer

Co-funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Cybersecurity Competence Centre. Neither the European Union nor the granting authority can be held responsible for them.

1.2 Distribution and License

The document is distributed under Creative Common Attribution 4.0 International [CC-BY](#).

The document is distributed as TLP:CLEAR.

1.3 Deliverable Definition

The identifier of the deliverable is D3.2 and it adheres to the definition outlined in the grant agreement **A list of commits of in public training material git repositories**. The deliverable name is **References of training material updates #2**.

1.4 Executive Summary

In the NGSOTI project, new training materials are developed, or existing ones are updated. Some of these materials are released under an open-source license, allowing multiple contributors from various projects to enhance and extend them. This report focuses on NGSOTI training programs in the following four domains:

- MISP
- AIL
- NEO-LEA
- NGSOTI (expert invited lectures)

In addition, Tenzir contributes cross-cutting training material on detection engineering, presented in a separate section at the end of this report.

This report includes references to commits in the public training material repositories. For the Tenzir material, the URLs are provided. The text in red includes hyperlinks that can be accessed online for verification. This deliverable is a follow-up to Deliverable D3.1. It covers the period from 1 April 2025 to 30 June 2026.

2 Training Material

2.1 MISP

The MISP open-source training materials have been developed over several years with support from multiple funding sources and external contributors. To track which parts were related to NGSOTI, the table below lists the open-source materials, along with their dates, training repository names, and commit IDs to retrieve the corresponding content. MISP plays a central role in NGSOTI by enabling integration with external sources. Throughout end of 2024 and early 2026, MISP training has been expanded to align with NGSOTI objectives. The commits listed under columns with the repository name 'misp-training' are taken from the [MISP-training](#) GitHub repository.

The list of commits related to trainings is shown in the table below including each commit reference in the GitHub link:

Date	Repository	Title	Commit
2025-04-16	misp-training	chg: [b.6-automation] improvements	#1
2025-04-25	misp-training	new: [events:FIRSTCTI25_MISP_automation_workshop] Added slides, custom modules and SkillAegis scenario	#2
2025-05-02	misp-training	new: [c.0-current-state] Added slide deck	#3
2025-05-05	misp-training	add: [complementary] Updated Data model overview in light color mode	#4
2025-05-26	misp-training	- chg [NATO MUG] presentation added	#5
2025-12-16	misp-training	Human Operated Ransomware pivot exercise	#6
2026-03-30	misp-training	new: [election-guidelines] First draft	#7
2026-03-30	misp-training	chg: [election-guidelines] Added AIL	#8
2026-03-30	misp-training	chg: [election-guideline] Added pdf	#9

2.2 AIL

AIL training material is maintained as an open-source resource and is regularly adjusted to reflect practical needs observed in information leak analysis, OSINT investigation, and threat intelligence workflows. For D3.2, the table below identifies the relevant updates supported by NGSOTI, with dates, repository references, short change descriptions, and commit links that allow the corresponding material to be retrieved. AIL supports NGSOTI activities by helping analysts collect, process, and pivot across information leaks and related datasets. Between 1st April 2025 and 30th June 2026, the training content was updated with revised introductions, new pivoting material, refreshed screenshots, migration work, and updated links. The commit references listed for the repository come from the public AIL-training GitHub repository [ail-training](#)

The list of commits related to trainings is shown in the table below, including each commit reference in the GitHub link:

Date	Repository	Title	Commit
2025-05-22	ail-training	chg: [short intro] update	#1
2025-05-22	ail-training	chg: [short intro] add search screenshot	#2
2025-05-22	ail-training	chg: [short] image missing	#3
2025-05-22	ail-training	chg: [training] ail-training updated	#4
2025-07-16	ail-training	chg: [short intro] migrate latex	#5
2025-07-16	ail-training	chg: [ail short intro] add ail internal	#6
2025-07-16	ail-training	chg: [ail short intro] remove old slide	#7
2025-07-16	ail-training	chg: [intro] migrate latex	#8
2025-07-16	ail-training	chg: [intro] improve feeders section	#9
2025-07-17	ail-training	new: [Art of Pivoting] New version added for VSS 2025 training	#10
2025-07-17	ail-training	chg: [short intro] add onion lookup + images descriptions	#11
2025-07-17	ail-training	chg: [short intro] Ongoing Developments	#12
2026-02-24	ail-training	chg: [short intro] update	#13
2026-04-21	ail-training	chg: [intro] update AIL intro	#14
2026-04-21	ail-training	chg: [intro] rename short intro to ail-introduction	#15

2.3 NEO-LEA updates

The **NEO-LEA** repository contains training material focusing on artefact analysis using forensic techniques used by SOC operators as well as law enforcement agencies. Analysis techniques for digital signatures have been added to strengthen capabilities for verifying and analysing digitally signed documents.

Date	Repository	Title	Commit
2026-05-29	e.200-dfir-pdf-analysis	added pdfsig tool to handle electronic signatures	#1
2026-05-29	e.200-dfir-pdf-analysis	added script to extract tool indicators that were used to sign	#2
2026-05-29	e.200-dfir-pdf-analysis	get indications on the tools that were used to sign	#3
2026-05-29	e.200-dfir-pdf-analysis	added example output of pdfsig	#4
2026-05-29	e.200-dfir-pdf-analysis	added example of openssl	#5

2.4 NGSOTI: Expert lectures

The **NGSOTI** repository hosts training materials developed within the project and contributed directly by the partners. Three modules were added covering complementary aspects of network security and digital trust infrastructure, and cybersecurity awareness through game-based learning. The first addresses DNS as an attack vector, covering look-alike domains, data exfiltration over DNS, and tunneling tools such as iodine and dnscat2, with hands-on detection using Wireshark, Snort, and Suricata signatures. The second examines long-term digital preservation from a cybersecurity perspective, grounding the topic in CEN/TS 18170:2025 and eIDAS 2: it covers the OAIS model (SIP/AIP/DIP), integrity chains, cryptographic audit trails, storage security requirements, and the implications of algorithm lifecycle management for archiving trust service providers. The third covers how to design educational games by defining a target audience, vulgarisation techniques such as analogy, storytelling, and visualisation, building a gameplay loop, choosing game elements, and running playtest

iterations, illustrated with a phishing-detection email exercise and a survey of existing cybersecurity games across board, digital, and simulation formats.

Date	Repository	Title	Commit
2026-06-16	training/UNILU-Network-Security	Add [network-security-dns] Network Security Wireshark	#1
2026-06-16	training/UNILU-Electronic-Archiving	Add [electronic-archiving] National Archives cybersecurity perspective	#2
2026-06-26	training/UNILU-NC3-Cybersecurity-and-Games	Add [cyber-and-games] NC3 CyberAndGames BenjaminJoly-OmarRamadan	#3

2.5 Tenzir

Tenzir contributes open-source training material on detection engineering, published as documentation at <https://docs.tenzir.com> (source repository: <https://github.com/tenzir/docs>). Within NGSOTI, Tenzir sits upstream of the SIEM to acquire, normalize, and route high-volume security telemetry, with a focus on analytical workloads such as detection engineering and threat hunting.

The material teaches SOC operators to express detections in the Tenzir Query Language (TQL) over real data. A key part is normalization: mapping raw events into common security data models such as OCSF, ECS, Microsoft Sentinel ASIM, Splunk CIM, and Google SecOps UDM, so that detections run against consistent, schema-aligned fields. On that normalized data, the material progresses from simple threshold rules to statistical anomaly detection. The relevant guides are:

- Map to OCSF — normalize events into the Open Cybersecurity Schema Framework: <https://docs.tenzir.com/guides/normalization/map-to-ocsf>
- Aggregate event streams — windowed threshold and statistical detections, including brute-force login detection and a traffic-spike detector (mean + 2 standard deviations) adapted from the open-source Splunk security_content project: <https://docs.tenzir.com/guides/analytics/aggregate-event-streams>
- Execute Sigma rules — running Sigma signatures on normalized telemetry: <https://docs.tenzir.com/guides/enrichment/execute-sigma-rules>
- Enrich with threat intelligence — intel-driven detection using lookup tables and indicators of compromise (e.g. from MISP): <https://docs.tenzir.com/guides/enrichment/enrich-with-threat-intel>

The Tenzir documentation has 1,867 commits in the reporting period, which would result in a table with 1,867 rows. For readability, this table was omitted from the report. The commits can be audited by consulting the following URL: [Tenzir docs commits, 1 April 2025 to 30 June 2026](#).